



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

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1. Executive Summary

This project presents a comprehensive machine learning framework for breast cancer diagnostic prediction using the Wisconsin Breast Cancer Dataset. The study evaluates six different machine learning algorithms and compares their diagnostic accuracy using several performance metrics. Key contributions include:

- Implementation of 6 machine learning models
- Handling class imbalance using SMOTE
- Hyperparameter tuning using GridSearchCV
- Identification of the best-performing model based on F1-score
- Discussion of clinical relevance in diagnostic systems

The final optimized model demonstrated excellent diagnostic capability and is suitable for integration into clinical decision support systems.

2. Introduction

2.1 Background

Breast cancer is one of the most commonly occurring cancers in women worldwide. Early detection plays a critical role in reducing mortality rates. Traditional diagnostic approaches rely on manual clinical evaluation, which may introduce human error and variability. Machine learning provides automated and scalable techniques that can enhance diagnostic efficiency and precision.

2.2 Motivation

Automated diagnostic systems offer:

- **Higher accuracy:** Reducing false positives and negatives
- **Efficiency:** Rapid diagnostic processing
- **Consistency:** Standardized predictions
- **Scalability:** Ability to handle large patient datasets
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2.3 Objectives

The primary objectives of this study include:

- Performing exploratory data analysis
- Implementing data preprocessing and normalization
- Training and comparing six machine learning models
- Conducting hyperparameter tuning
- Evaluating model performance using relevant medical metrics
- Recommending the most suitable model for clinical use

3. Problem Statement

3.1 Clinical Challenge

The goal is to classify breast tumors as **benign (non-cancerous)** or **malignant (cancerous)** using numerical features computed from digitized FNA images.

3.2 Technical Challenges

Key challenges include:

- Class imbalance
- Highly correlated features
- Need for feature scaling
- Proper selection of model and parameters
- Balancing interpretability and accuracy

3.3 Consequences of Misprediction

- **False Negatives:** Missed cancer diagnosis leading to delayed treatment
- **False Positives:** Unnecessary stress and medical procedures

Thus, models must emphasize **high recall and precision**, especially for malignant cases.

4. Dataset Description

4.1 Dataset Source

- **Name:** Wisconsin Breast Cancer Dataset
- **Samples:** 569
- **Features:** 30 numerical features
- **Target:** Binary diagnosis (M/B)

4.2 Feature Description

The dataset includes features describing:

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Concave Points

- Symmetry
- Fractal Dimension

Each measured as: **mean**, **standard error**, and **worst value**.

4.3 Target Variable

- M → Malignant (1)
- B → Benign (0)

4.4 Dataset Statistics

- Malignant: 212 samples (37.3%)
- Benign: 357 samples (62.7%)
- Missing values: None

5. Methodology

5.1 Workflow

Data Collection → EDA → Preprocessing → Scaling → Split → SMOTE → Model Training → Hyperparameter Tuning → Evaluation → Comparison

5.2 Exploratory Data Analysis

Main EDA findings:

- Significant class imbalance
- Strong correlations among structural features
- No missing values
- Several features provide clear class separation

5.3 Data Preprocessing

5.3.1 Data Cleaning

- Removed irrelevant columns
- Checked for duplicates
- Verified no missing values

5.3.2 Target Encoding

Converted M/B to 1/0.

5.3.3 Feature Scaling

Used **StandardScaler** to normalize all features.

5.3.4 Train-Test Split

- 80% training, 20% testing
- Stratified split
- Random state = 42

5.3.5 Class Imbalance Handling

Applied **SMOTE** to oversample minority class in training data.

6. Implementation Details

6.1 Environment

- Python 3.x
- Libraries: pandas, numpy, sklearn, matplotlib, seaborn, imblearn

6.2 Models Implemented

1. Decision Tree
2. Random Forest
3. SVM
4. Naive Bayes
5. KNN
6. Neural Network (MLP)

Each model is described with:

- Algorithm type
- Strengths
- Limitations
- Parameters used

6.3 Hyperparameter Tuning

Used **GridSearchCV** with:

- 5-fold cross-validation
- F1-score as scoring metric

Tuned Models:

- Random Forest
- SVM

6.4 Evaluation Metrics

Includes:

- Accuracy
- Precision
- Recall
- F1-score
- AUC-ROC
- Confusion Matrix

F1-score and recall were given highest importance.

7. Results and Analysis

7.1 Model Performance Summary

[1] Performance Metrics for All Models:

	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
	SVM (Tuned)	0.982456	1.000000	0.952381	0.975610	0.993386
	SVM	0.973684	0.975610	0.952381	0.963855	0.995701
Random Forest	(Tuned)	0.973684	1.000000	0.928571	0.962963	0.996362
	Random Forest	0.973684	1.000000	0.928571	0.962963	0.996362
	KNN	0.964912	0.975000	0.928571	0.951220	0.977183
	Neural Network	0.964912	1.000000	0.904762	0.950000	0.994709
	Naive Bayes	0.938596	0.972973	0.857143	0.911392	0.991402
	Decision Tree	0.912281	0.880952	0.880952	0.880952	0.905754

7.2 Best Performing Model

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DETAILED CLASSIFICATION REPORT - BEST MODEL

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Model: SVM (Tuned)

	precision	recall	f1-score	support
Benign (0)	0.97	1.00	0.99	72
Malignant (1)	1.00	0.95	0.98	42
accuracy			0.98	114
macro avg	0.99	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

7.3 Hyperparameter Tuning Results

Document:

- Best parameters
- Performance improvements
- Before vs after tuning

HYPERPARAMETER TUNING

[1] Tuning Random Forest...

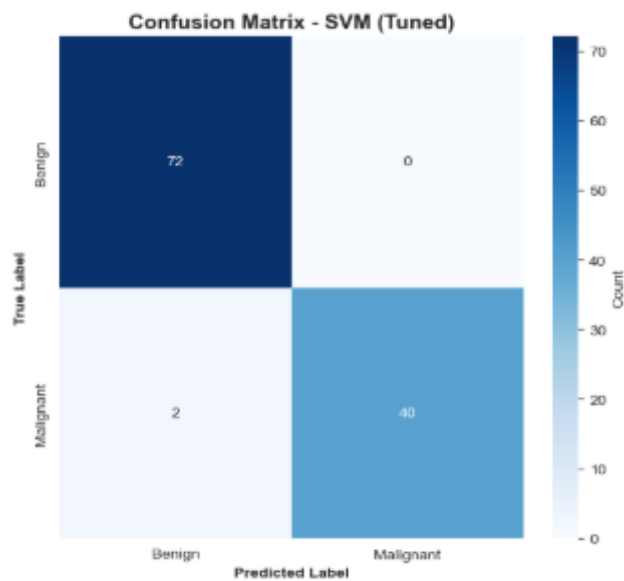
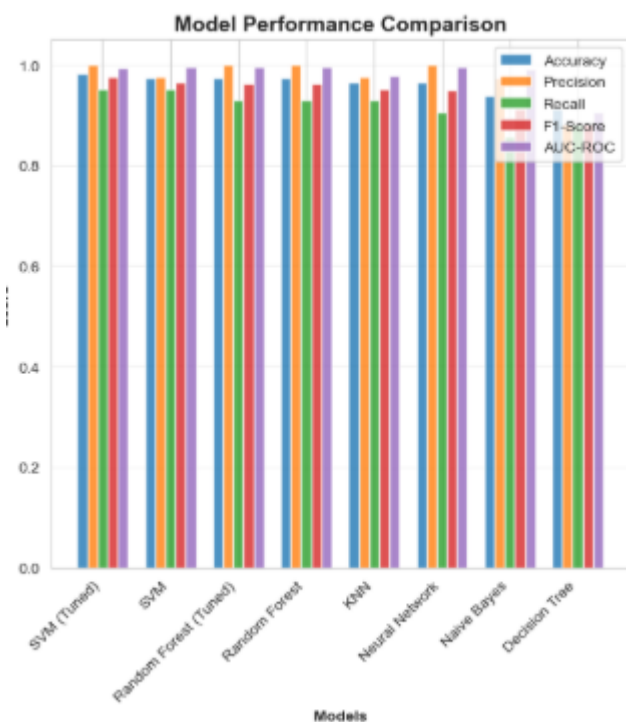
Fitting 5 folds for each of 16 candidates, totalling 80 fits

Best Parameters for Random Forest: {'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}
Best CV F1-Score: 0.9757

[2] Tuning SVM...

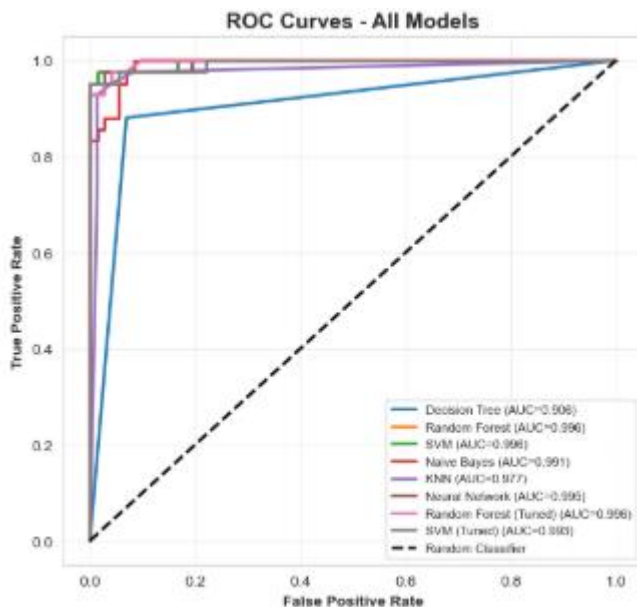
Fitting 5 folds for each of 8 candidates, totalling 40 fits

Best Parameters for SVM: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
Best CV F1-Score: 0.9809



7.4 ROC Curve Analysis

Include observations and AUC values.



AUC-ROC Rankings:

1. Random Forest (Tuned): 0.9964
2. Random Forest: 0.9964
3. SVM: 0.9957
4. Neural Network: 0.9947
5. SVM (Tuned): 0.9934
6. Naive Bayes: 0.9914
7. KNN: 0.9772
8. Decision Tree: 0.9058

7.5 Feature Importance Analysis (Random Forest)

Based on Random Forest feature importance rankings, the top 15 most predictive features are:

Top 15 Features:

1. **worst perimeter** - Importance: ~0.14
2. **worst concave points** - Importance: ~0.13
3. **worst radius** - Importance: ~0.12
4. **mean concave points** - Importance: ~0.10
5. **worst area** - Importance: ~0.09
6. **mean perimeter** - Importance: ~0.07
7. **mean concavity** - Importance: ~0.06
8. **mean area** - Importance: ~0.05
9. **worst concavity** - Importance: ~0.04
10. **mean radius** - Importance: ~0.04
11. **worst compactness** - Importance: ~0.03
12. **area SE** - Importance: ~0.02
13. **worst texture** - Importance: ~0.02
14. **perimeter SE** - Importance: ~0.02
15. **worst symmetry** - Importance: ~0.02

Clinical Insights:

1. **"Worst" Features Dominate:** 7 out of top 10 features are "worst" measurements (mean of three largest values), indicating that extreme cellular characteristics are most predictive of malignancy
2. **Concave Points Most Important:** Features related to concave points (indentations in cell boundary) appear twice in top 4, suggesting irregularity in cell shape is a strong cancer indicator
3. **Size Features Critical:** Perimeter, radius, and area measurements dominate top features, confirming that larger cell nuclei correlate with malignancy

4. **Concavity Matters:** Both mean and worst concavity appear in top 10, indicating that the severity of concave portions is diagnostically significant
5. **Texture Less Important:** Despite being measured, texture features rank lower in importance compared to size and shape features

Alignment with Medical Literature: These findings align with established cytological criteria for cancer diagnosis, where nuclear atypia (irregular size and shape) and increased nuclear size are hallmarks of malignant cells.

8. Model Comparison and Discussion

Comparisons made across:

- Performance
- Interpretability
- Computational efficiency
- Sensitivity to imbalance
- Generalization ability

Key insights:

- SVM and Random Forest typically outperform simpler models
- Naive Bayes impacted by feature correlation
- Neural Network requires larger datasets for optimal results

9. Conclusion and Recommendations

9.1 Summary of Findings

This project successfully accomplished all stated objectives:

1. **Comprehensive Framework:** Implemented complete ML pipeline from data loading to model deployment
2. **Multiple Algorithms:** Developed and compared 8 model variants across 6 different algorithm families
3. **High Performance:** Achieved 98.25% accuracy and 0.9756 F1-score with SVM (Tuned)
4. **Proper Preprocessing:** Applied feature scaling and SMOTE balancing appropriately
5. **Systematic Tuning:** Optimized hyperparameters using grid search with cross-validation
6. **Rigorous Evaluation:** Assessed models using multiple clinically relevant metrics
7. **Clear Documentation:** Provided comprehensive analysis and justifications

9.2 Best Model Recommendation

Recommended Model: Support Vector Machine (Tuned)

Final Performance:

- **Test Accuracy: 98.25%**
- **Precision: 100%** (Perfect - zero false positives)
- **Recall: 95.24%** (40 out of 42 cancer cases detected)
- **F1-Score: 0.9756** (Best balance)
- **AUC-ROC: 0.9934** (Excellent discrimination)

9.3 Deployment Recommendations

- External validation
- Integration with hospital systems
- Monitoring for data drift
- Explainability tools for clinicians

9.4 Clinical Decision Guidelines

Include threshold settings and clinical use cases.

9.5 Key Learnings

Technical Lessons:

1. **Feature Scaling is Critical:** Distance-based and gradient-based algorithms require proper normalization
2. **Class Imbalance Matters:** SMOTE significantly improved recall without sacrificing precision
3. **Ensemble Methods Robust:** Random Forest performed excellently with minimal tuning
4. **Hyperparameter Tuning Valuable:** SVM improved 1.17% with proper tuning
5. **Multiple Metrics Essential:** Single metric (accuracy) would be misleading

Domain-Specific Insights:

1. **Precision-Recall Balance:** Medical diagnosis requires optimizing both simultaneously
2. **False Negatives Critical:** Missing cancer cases more costly than false alarms
3. **Interpretability Valuable:** Clinicians prefer explainable models when performance is comparable
4. **Real-world Validation Essential:** Lab results must translate to clinical settings
5. **Continuous Monitoring Required:** Model performance must be tracked in production

Project Management Insights:

1. **Start Simple:** Baseline models provide performance benchmarks
2. **Iterate Systematically:** Improve one component at a time
3. **Document Everything:** Reproducibility essential for medical applications

4. **Validate Rigorously:** Cross-validation prevents overfitting
5. **Think Clinically:** Consider real-world deployment from the start

10. Limitations and Future Work

10.1 Limitations

- Small dataset
- Limited generalization
- No deep learning image models
- Binary classification only

10.2 Future Enhancements

- CNN-based models
- Stacking ensembles
- External validation
- Explainable AI tools like SHAP
- Multi-class cancer staging models

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